

CSE36301 Machine Learning Assignment1

Deadline: Sunday, September 28, 2025, 11:59 PM

1 Linear Algebra

1.1 Matrix Operations

Consider the following matrices over $\mathbb{R}$:

$$A = \begin{bmatrix} 1.3443 & -2.3342 & 4.5461 \\ -2.3234 & 3.4535 & -5.6617 \end{bmatrix}, \quad (1)$$

$$B = \begin{bmatrix} 2.3724 & 4.5212 \\ 3.4564 & 6.7448 \\ 1.2643 & -2.3254 \end{bmatrix}, \quad (2)$$

$$C = \begin{bmatrix} -1.2373 & 1.2334 \\ 3.3122 & 3.2122 \end{bmatrix} \quad (3)$$

Task: Determine, if possible, the following matrix products:

- CA
- BC
- $B^T A$
- $A^T C$
- $(-A)^T C$
- $B^T A^T$
- AC
- CB

If any computation is impossible, explain why based on matrix dimension compatibility.

Instructions: You may use calculators for computations. You don't need to show the detailed calculation steps. Compute with full precision, but round your final answers to 3 decimal places.

1.2 Properties of Determinants

Task: Construct matrices $A, B \in \mathbb{R}^{n \times n}$ such that $\det(A + B) \neq \det(A) + \det(B)$ for:

- (a) $n = 2$
- (b) $n = 3$
- (c) $n = 4$

For each case, provide specific matrices and verify your result by computing the determinants.

- (d) Can you find a general pattern or construction method that works for any n ?

Instructions: You may use calculators for determinant computations. You don't need to show detailed calculation steps, but your equations and final results must be accurate.

1.3 Computing Eigenvalues and Eigenvectors

Task: Determine the **eigenvalues** and corresponding **eigenvectors** of the following matrices over $\mathbb{R}$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad (4)$$

$$B = \begin{bmatrix} 4 & -12 & 5 \\ 0 & 2 & 13 \\ 0 & 3 & -1 \end{bmatrix}, \quad (5)$$

$$C = \begin{bmatrix} 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & -2 \end{bmatrix} \quad (6)$$

Additional Question: Is there any difference when you consider A , B , and C as matrices over $\mathbb{C}$ instead of $\mathbb{R}$?

2 Linear Regression

In this section, you will implement linear regression from scratch using two different approaches: the analytical normal equation method and the iterative gradient descent optimization. Start your work from attached `Assignment1.ipynb` file.

2.1 Normal Equation Method

The normal equation provides a closed-form analytical solution to linear regression by directly computing the optimal parameters θ that minimize the mean squared error. This approach solves the optimization problem algebraically without requiring iterative optimization.

Task: In your `.ipynb` file, complete the TODO sections in the following functions:

- `add_intercept`
- `normal_equation_solution`
- `compute_mse`
- `predict`

Validation: Run `test_with_diabetes_dataset()` in the next cell to validate your implementation. If your implementation is correct, the results should be similar to the Scikit-learn implementation.

2.2 Gradient Descent Method

Gradient descent implements an iterative optimization approach to find the optimal parameters by following the negative gradient of the cost function. This method often scales better to large datasets compared to the normal equation method.

Task: In your `.ipynb` file, complete the TODO sections in the following functions:

- `compute_cost_and_gradient`
- `gradient_descent`

Validation: Run `test_gradient_descent_convergence` in the next cell to validate your implementation. If your implementation is correct, the results should demonstrate acceptable convergence. You may need to adjust the learning rate if convergence is not observed.

3 Submission Requirements

Your submission file must be a .zip file with:

- `Assignment1_[YOUR_STUDENT_ID]_Linear_Algebra.[YOUR_FILE_FORMAT]` (for Linear Algebra)
- `Assignment1_[YOUR_STUDENT_ID]_Linear_Regression.ipynb` (for Linear Regression Code)
- `Assignment1_[YOUR_STUDENT_ID]_Linear_Regression.pdf` (for Linear Regression Outputs)

Example:

- `Assignment1_20241234_Linear_Algebra.pdf` (for Linear Algebra)
- `Assignment1_20241234_Linear_Regression.ipynb` (for Linear Regression Code)
- `Assignment1_20241234_Linear_Regression.pdf` (for Linear Regression Outputs)

3.1 Linear Algebra Section

Any submission format is acceptable (handwritten, tablet, $\text{\LaTeX}$, etc.). If you are handwriting your answers, while the TA will make every effort to read handwritten solutions, please try to write neatly to avoid situations where your work cannot be recognized.

3.2 Linear Regression Section

Submit both:

1. Your completed .ipynb file
2. A .pdf file containing the printed output of your notebook

4 Contact

If you have any questions regarding this assignment, please email the TAs:

- Seokhoon Jeong: `shjd0246@unist.ac.kr`
- Jinsik Bang: `bang@unist.ac.kr`